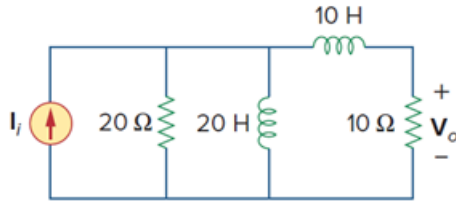


Problem

For the circuit shown in Fig. 14.70, find $\mathbf{H}(s) = \mathbf{V}_o(s)/\mathbf{I}_i(s)$.

Figure 14.70:



Step-by-step solution

Step 1 of 3

Refer to circuit diagram Figure 14.70 in the text book.

In frequency domain inductor is represented as sL .

Draw the frequency domain circuit.

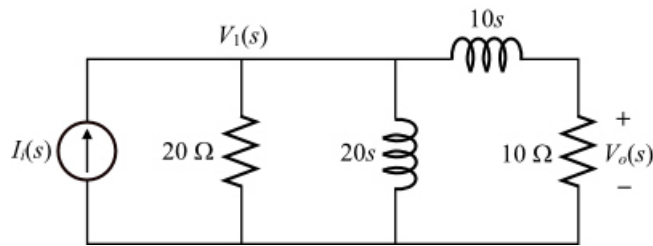


Figure 1

Step 2 of 3

Apply Kirchhoff's current law at node $V_1(s)$.

$$\begin{aligned} \frac{V_1(s)}{20} + \frac{V_1(s)}{20s} + \frac{V_1(s)}{10+10s} &= I_i(s) \\ \left(\frac{1}{20} + \frac{1}{20s} + \frac{1}{10+10s} \right) V_1(s) &= I_i(s) \\ \left(\frac{s(10+10s) + 10 + 10s + 20s}{(20s)(10+10s)} \right) V_1(s) &= I_i(s) \\ \left(\frac{10s + 10s^2 + 10 + 10s + 20s}{(20s)(10+10s)} \right) V_1(s) &= I_i(s) \\ V_1(s) &= \left(\frac{(2s)(10+10s)}{s^2 + 4s + 1} \right) I_i(s) \dots\dots (1) \end{aligned}$$

Step 3 of 3

Calculate the transfer function using voltage division rule.

$$V_o(s) = \left(\frac{10}{10s+10} \right) V_1(s)$$

Substitute $\left(\frac{(2s)(10+10s)}{s^2+4s+1} \right) I_i(s)$ for $V_1(s)$.

$$\begin{aligned} V_o(s) &= \left(\frac{10}{10s+10} \right) \left(\frac{(2s)(10+10s)}{s^2+4s+1} \right) I_i(s) \\ &= \left(\frac{20s}{s^2+4s+1} \right) I_i(s) \end{aligned}$$

$$\frac{V_o(s)}{I_i(s)} = \frac{20s}{s^2+4s+1}$$

Therefore, the transfer function, $\frac{V_o(s)}{I_i(s)}$ of the circuit is $\boxed{\frac{20s}{s^2+4s+1}}$.